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Bi-directional attention network with drop aggregation for microRNA-disease association prediction

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ABSTRACT

Identifying the potential relationships between microRNAs (miRNAs) and human diseases is vital for advancing our knowledge of pathological mechanisms and promoting therapeutic development. Nevertheless, many current computational models make insufficient use of heterogeneous biological resources and depend on simplistic feature extraction, which limits their ability to capture the intricate non-linear associations between miRNAs and diseases. To overcome these challenges, we present BADMDA, an innovative framework that combines centered kernel alignment-based multiple kernel learning (CKA-MKL), drop aggregation (DropAGG), and a bi-directional attention mechanism. In this design, CKA-MKL adaptively assigns weights to different similarity kernels, thereby achieving an effective integration of diverse information sources. Based on established miRNA-disease associations (MDAs), we then build a bipartite attributed graph, where DropAGG learns expressive multi-level features and alleviates over-smoothing. The bi-directional attention component further improves feature quality by capturing reciprocal dependencies between diseases and miRNAs. A multi-layer perceptron is finally employed to infer novel associations from the enhanced representations. The superior accuracy of BADMDA is attributed to its ability to adaptively fuse heterogeneous similarities and retain complex non-linear dependencies through attention-driven feature learning. On HMDD v2.0 and HMDD v3.2, it achieves AUC values of 0.9372 and 0.9533 and AUPR values of 0.9348 and 0.9525, respectively, surpassing ten advanced methods. Case studies on lung neoplasms and hepatocellular carcinoma further confirm the robustness and biological relevance of BADMDA.

1. Introduction

With the swift progress of genomics and bioinformatics, the pivotal role of noncoding RNAs (ncRNAs), especially microRNAs (miRNAs), in gene regulation has been increasingly acknowledged (Zhao et al., 2025). Comprising 21–25 nucleotides, miRNAs regulate gene expression during the phenomena of cell proliferation events, the differentiation of cells, and the occurrence of apoptosis. Malfunctions in miRNA expression regulation are closely linked to cancer (Vedanayagam et al., 2019), cardiovascular diseases (Laggerbauer & Engelhardt, 2022), and neurological disorders (Alkhazaali-Ali et al., 2024). For instance, miR-107 exerts neuroprotective effects through multiple pathways in Alzheimer's disease, regulating aspects such as A β production, tau protein phosphorylation, and the function and survival of neurons (Zhou et al., 2023). miR-1 regulates HDAC4 to promote cardiac cell differentiation and sur-

vival, thereby reducing cell death after myocardial infarction (Huang & Wang, 2024). In breast cancer, miR-145 reduces cell viability by regulating specific genes. miR-145 downregulates c-Myc to impede the growth of non-small cell lung cancer cells (Piao et al., 2022). Therefore, pinpointing miRNA-disease associations (MDAs) is the cornerstone of comprehending pathogenesis and discovering biomarkers. However, traditional wet-laboratory for identifying MDAs are prolonged and exorbitant (Aguilera et al., 2023). Consequently, a multitude of computational approaches have been devised as efficient instruments for forecasting potential MDAs (Górriz et al., 2023). These strategies are sorted into three specific types: information dissemination-based, scoring function-based, and deep learning-based approaches.

Under information dissemination-oriented strategies, the anticipation of most disease-associated miRNAs is achieved through the integration of various resources. He (Ha, 2023) introduced a similarity-based

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matrix factorization approach for predicting MDAs. MLPMDA model (Guo et al., 2021) integrates multi-layer linear projection to estimate MDA scores. Yuan et al. (2024) employed a heterogeneous biological network consisting of diseases, drugs, and ncRNAs to infer unknown linkages. Liu et al. (2024) predicted drug-disease associations (DDAs) by constructing a dual graph transformer network and using multi-modal information to capture complex relationships and latent features. Yu et al. (2022) adopted MDPBMP, which leverages seven meta-paths in a heterogeneous network for MDA prediction. Wang et al. (2023) utilized MSGCA with multi-similarities graph convolutional autoencoder (AE) to anticipate drug-disease undiscovered relationships.

Methods relying on scoring functions generally rest on specific pre-suppositions. The likelihood of the MDAs is computed by reasonably exploiting diverse similarity metrics. Chen et al. (2018) designed EGBM-MDA to forecast MDAs, which integrates known MDAs, miRNA functional similarity and disease semantic similarity matrices (DSSM). Bi-SGTAR (Li et al., 2024) was proposed, which employs an encoder with sparse gating to evaluate circRNA-disease associations (CDAs). Dai et al. (2021) came up with LWBRW approach in order to estimate underlying connections. GCDSAEMDA (Wang et al., 2025) utilized heterogeneous graph attention network and deep sparse autoencoder to calculate microbe-disease association scores. Gu and Li (2023) incorporated multi-classifiers into the ELMDA model to reveal potential correlations.

With deep learning's wide application across domains, researchers have applied it to predict MDAs. Guo et al. (2023) utilized multi-view feature-driven variational gated AE model for feature extraction in disease-miRNA association inference. Ha and Park (2022) made use of a node2vec-oriented neural collaborative filtering method to detect MDAs. Sun et al. (2025) presented K-MGCMLD method, which is aimed at anticipating linkages of diseases, lncRNAs, and miRNAs. Kang et al. (2024) exploited a kernel fusion and deep auto-encoder framework called KFDAE for predicting CDA. HyperGRL-CDA (Li et al., 2025) makes use of hypergraph representation learning to detect CDAs. Liu et al. (2022) devised GCNPCA utilizing graph convolutional network (GCN). Long et al. (2024) applied the TriFusion model, which uses a tri-channel fusion neural network to anticipate MDAs. MSMCDA (Zhang et al., 2025) integrates multi-view shared units and multi-channel attention, using contrastive learning to boost CDA prediction. CircDA (Niu et al., 2024) is a CDA-focused algorithm built on a Tumor-based deep learning framework, whose core mechanism abstracts biomolecules as nodes and intermolecular interactions as edges, and integrates multi-omics data to construct a heterogeneous biomolecular association network.

Nevertheless, many current computational models attempt to utilize the diverse biological information available, yet the use of relative feature extraction strategies often limits their ability to accurately model the complex, non-linear relationships between miRNAs and their associated diseases. To address these limitations, we propose BADMDA, an innovative framework that combines centered kernel alignment-based multi-kernel learning (CKA-MKL) for adaptive multi-source similarity integration, drop aggregation (DropAGG) for robust multi-level feature extraction, and bi-directional attention for capturing reciprocal miRNA-disease dependencies, enabling more accurate prediction of novel associations.

Overall, BADMDA contributes in three primary contributions:

- CKA-MKL adaptively integrates multiple sources of prior knowledge, effectively capturing complementary information, reducing redundancy, and enabling the model to exploit heterogeneous similarity data for improved feature representation.
- DropAGG is designed as a generalized scheme that can be incorporated into the bi-directional attention network to extract robust multi-level features while mitigating over-smoothing across the graph.
- The bi-directional attention module influences flowing in both directions between miRNAs and diseases, simultaneously reducing dimen-

Table 1

Description of parameter and abbreviation.

Parameter/Abbreviation	Description
ncRNAs	noncoding RNAs
miRNAs	microRNAs
MDAs	miRNA-disease associations
CDAs	circRNA-disease associations
DDAs	drug-disease associations
AE	autoencoder
VAE	variational autoencoders
DSSMs	disease semantic similarity matrices
GCN	graph convolutional network
GNNs	graph neural networks
CKA-MKL	centered kernel alignment-based multi-kernel learning
DropAGG	drop aggregation
HMDD	Human microRNA Disease Database
GIP	gaussian interaction profile
5-CV	5-fold cross-validation
AUC	Area Under the Receiver Operating Characteristic Curve
AUPR	Area Under the Precision-Recall Curve
ACC	accuracy
F1	F1-score
PRE	precision
n	training epochs
lr	learning rate
d	feature dimension
l	number of convolution layer

sionality, enhancing feature discriminability, and capturing complex non-linear dependencies.

2. Materials and methods

This section describes the implementation of the BADMDA model, and Fig. 1 illustrates it visually. (A) Data Preprocessing and Multi-Kernel Similarity Fusion: By means of CKA-MKL, ID and IM for disease and miRNA integrated similarity matrix are produced. (B) Bi-directional Attention Network: A bipartite attributed graph is built using known MDAs, along with the ID and IM . DropAGG extracts hierarchical features, which are then refined by a bi-directional attention module to enhance feature representation. (C) Inferring MDAs: The final embeddings are input into a multi-layer perceptron to infer MDAs.

2.1. Datasets

For evaluating the BADMDA model's generalization capability, two editions of the Human microRNA Disease Database (HMDD) were implemented. HMDD v2.0 (Li et al., 2014) contains 383 diseases, 495 miRNAs, and 5430 are relationships, while HMDD v3.2 (Huang et al., 2019) includes 792 diseases, 917 miRNAs, and 14550 correlations. Two datasets are utilized to construct MDA matrix $A \in \{0, 1\}^{N \times D}$, where N and D signify the quantity of miRNAs and diseases, in respective order.

2.2. Data preprocessing

2.2.1. Preliminary

Disease semantic similarity: $DSSM_1$ (Wang et al., 2010) and $DSSM_2$ (Xuan et al., 2013), two disease semantic similarity matrices (DSSMs) are formulated according to different criteria. The final $DSSM$ (denoted as D_{ss}) was obtained as follows:

$$D_{ss} = \frac{DSS_1 + DSS_2}{2}, \quad (1)$$

Gaussian interaction profile (GIP) kernel similarity: Following preceding research (Van Laarhoven et al., 2011), the interaction characteristics of disease d_i (miRNA m_i) were represented by the vector $V(d_i)(V(m_i))$, capturing its associations with various miRNAs (diseases)

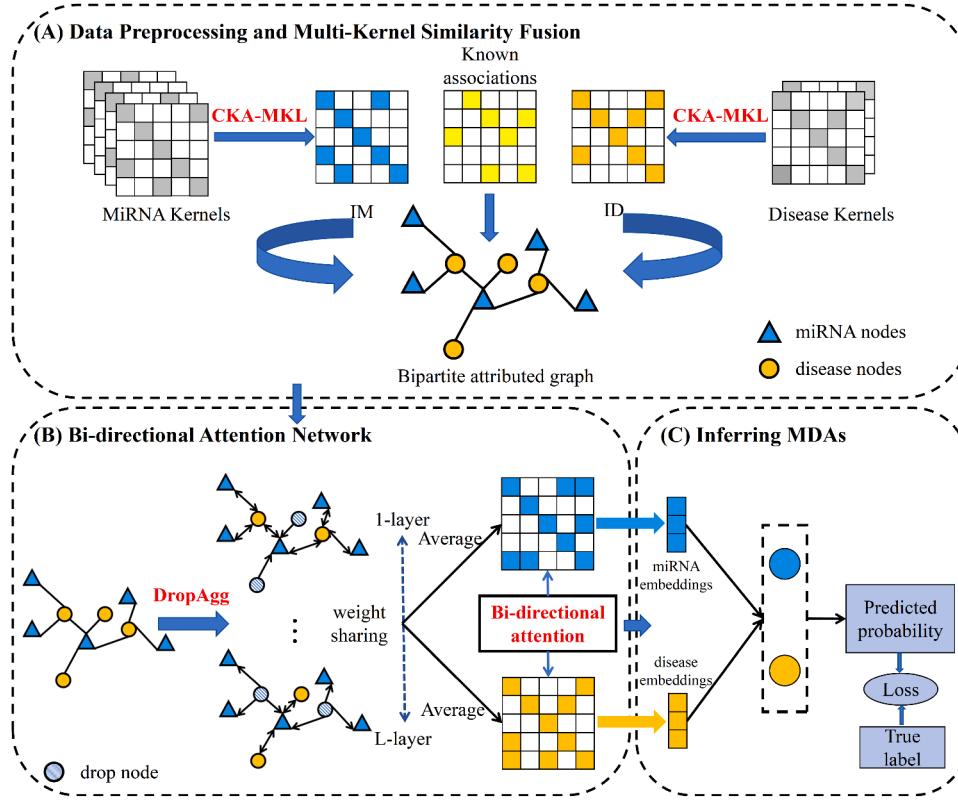


Fig. 1. The framework of BADMDA.

as follows:

$$\begin{aligned} D_{gip}(d_i, d_j) &= \exp(-\lambda_d \|\mathbf{V}(d_i) - \mathbf{V}(d_j)\|^2), \\ M_{gip}(m_i, m_j) &= \exp(-\lambda_m \|\mathbf{V}(m_i) - \mathbf{V}(m_j)\|^2), \\ \lambda_d &= 1 / \frac{1}{ND} \sum_{i=1}^{ND} \|\mathbf{V}(d_i)\|^2, \lambda_m = 1 / \frac{1}{NM} \sum_{i=1}^{NM} \|\mathbf{V}(m_i)\|^2, \end{aligned} \quad (2)$$

where λ_d and λ_m act as kernel bandwidth parameters.

MiRNA family similarity: MiRNAs within the same family typically show structural and functional similarities from related biological processes. Based on the miRBase (Griffiths-Jones, 2010) database, miRNAs in the same family were scored as 1, and otherwise scored as 0 to define miRNA family similarity.

$$M_{fam} = \begin{cases} 1, m_i \text{ and } m_j \text{ belong to the same family} \\ 0, \text{otherwise} \end{cases}, \quad (3)$$

MiRNA functional similarity: MiRNAs act via target genes, so their functional similarity can be estimated by evaluating the similarity between their respective gene sets using the following equation:

$$M_{fun}(m_i, m_j) = \frac{\sum_{1 \leq i \leq N_i} \max(S_i^*) + \sum_{1 \leq j \leq N_j} \max(S_j^*)}{N_i + N_j}, \quad (4)$$

where S_i^* and S_j^* in the context of the target sets are taken to signify the functional similarities of gene i and gene j , individually. Meanwhile, N_i and N_j are utilized to denote the amounts of target genes corresponding to miRNA m_i and m_j , severally.

MiRNA sequence similarity: Levenshtein distance, which determines the minimum atomic character modifications (additions, removals, replacements) to convert one sequence to another, is used to compute sequence similarity. This distance is normalized as detailed below:

$$M_{seq}(m_i, m_j) = 1 - \frac{\text{Lev}(m_i, m_j)}{\max(|m_i|, |m_j|)}, \quad (5)$$

where $|m_i|$ and $|m_j|$ denote the sequence lengths.

MiRNA integrated similarity: miRNA family similarity, functional similarity, and sequence similarity of miRNAs are first initially integrated and defined as follows:

$$M_{int}(m_i, m_j) = \begin{cases} \theta_1 M_{fun} + (1 - \theta_1) M_{seq}, & \text{if } M_{fam}(m_i, m_j) = 0 \\ \theta_2(\theta_1 M_{fun} + (1 - \theta_1) M_{fam}) + (1 - \theta_2) M_{seq}, & \text{otherwise} \end{cases}, \quad (6)$$

where parameters θ_1 and θ_2 balance the similarity matrices, and are both set to 0.5.

2.2.2. CKA-MKL

CKA-MKL is implemented to adaptively blend various forms of prior knowledge stemming from the feature spaces of miRNAs and diseases. By using CKA-MKL, it becomes possible to efficiently seize complementary information and concurrently cut down on redundancy. Let $M = \{M_{int}, M_{gip}\}$ (redefine as $M = \{M_1, M_2\}$) and $D = \{D_{ss}, D_{gip}\}$ (redefine as $D = \{D_1, D_2\}$), in turn, stand for distinct kernels originating from miRNA and disease spaces, respectively. The optimal kernels IM and ID , are computed as follows:

$$\begin{aligned} IM &= \sum_{i=1}^{N_m} \omega_m^i M_i, M_i \in \mathbb{R}^{NM \times NM} \\ ID &= \sum_{j=1}^{N_d} \omega_d^j D_j, D_j \in \mathbb{R}^{ND \times ND} \end{aligned} \quad (7)$$

where ω_m^i and ω_d^j are considered the optimal weights. And the quantities of kernels allocated to miRNAs and diseases are denoted by N_m and N_d , individually.

The training set is employed by the CKA-MKL to figure out the optimal weights for all kernels. The kernel alignment score is characterized

in the following manner:

$$A(\mathbf{I}, \mathbf{J}) = \frac{\langle \mathbf{I}, \mathbf{J} \rangle_F}{\|\mathbf{I}\|_F \|\mathbf{J}\|_F} \quad (8)$$

where the Frobenius inner product is signified as $\langle \mathbf{I}, \mathbf{J} \rangle_F = \text{Trace}(\mathbf{I}^T \mathbf{J})$, and $\|\mathbf{I}\|_F = \sqrt{\langle \mathbf{I}, \mathbf{I} \rangle_F}$ denotes the Frobenius norm.

The kernel alignment score is capable of reflecting the bonds between the feature kernel $\mathbf{M}_i$ and the ideal kernel $\mathbf{M}_{ideal} \mathbf{M}_{ideal}^T$. By setting all the labels in test set to zero, the labels of the training set can be inferred.

$$\omega_m^i = \frac{A(\mathbf{M}_i, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T)}{\sum_{o=1}^{N_m} A(\mathbf{M}_o, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T)}, \quad (9)$$

In order to acquire the optimal kernel weights, the alignment score needs to be maximized. This is accomplished through the optimization of the following function:

$$\begin{aligned} \max_{\omega_m^i \geq 0} C A(\mathbf{I}\mathbf{M}, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T) \\ = \max_{\omega_m^i \geq 0} \frac{\langle \mathbf{C}_n \mathbf{I}\mathbf{M} \mathbf{C}_n, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T \rangle_F}{\|\mathbf{C}_n \mathbf{I}\mathbf{M} \mathbf{C}_n\|_F \|\mathbf{M}_{ideal} \mathbf{M}_{ideal}^T\|_F}, \end{aligned} \quad (10)$$

where $\mathbf{C}_n \in \mathbb{R}^{N \times N \times M}$ represents the centering matrix, and it symbolizes the average operator.

And Eq. (10) can be rewritten in the following manner:

$$\max_{\omega_m^i \geq 0} C A(\mathbf{I}\mathbf{M}, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T) = \max_{\omega_m^i \geq 0} \frac{\langle \mathbf{C}_n \mathbf{I}\mathbf{M} \mathbf{C}_n, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T \rangle_F}{\|\mathbf{C}_n \mathbf{I}\mathbf{M} \mathbf{C}_n\|_F} \quad (11)$$

The Eq. (11) can be interpreted in the following manner:

$$\max_{\omega_m^i \geq 0} \frac{\omega_m^T b}{\sqrt{\omega_m^T \mathbf{M} \omega_m}} \quad (12)$$

where the vector $\langle \mathbf{C}_n \mathbf{M}_i \mathbf{C}_n, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T \rangle_F, \dots, \langle \mathbf{C}_n \mathbf{M}_{N_m} \mathbf{C}_n, \mathbf{M}_{ideal} \mathbf{M}_{ideal}^T \rangle_F^T$ is denoted as b , and $\mathbf{M}$ is the following matrix:

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{1,1} & \mathbf{M}_{1,2} & \dots & \mathbf{M}_{1,N_m} \\ \mathbf{M}_{2,1} & \mathbf{M}_{2,2} & \dots & \mathbf{M}_{2,N_m} \\ \vdots & \vdots & \mathbf{M}_{e,f} & \vdots \\ \mathbf{M}_{N_m,1} & \mathbf{M}_{N_m,2} & \dots & \mathbf{M}_{N_m,N_m} \end{bmatrix}_{N_m \times N_m}, \quad (13)$$

where $\mathbf{C}_{e,f} = \langle \mathbf{C}_n \mathbf{M}_e \mathbf{C}_n, \mathbf{C}_n \mathbf{M}_f \mathbf{C}_n \rangle (e, f = 1, 2, \dots, N_m)$.

Subsequently, solving a quadratic programming problem allows us to obtain the solution to Eq. (10):

$$\min_{\omega_m^i \geq 0} \omega_m^T \mathbf{M} \omega_m - 2\omega_m^T b. \quad (14)$$

In a similar manner, the weights for the disease similarity kernel can as well be estimated by means of the previously mentioned CKA-MKL approach. Moderately increasing the regularization strength can enhance the robustness of the obtained weights.

2.2.3. Construct bipartite attributed graph

The bipartite attributed graph consists of known MDAs matrix A , attribute matrix X_m obtained from IM , and X_d obtained from ID processed by CKA-MKL. Consider an input graph $\tilde{G} = (V, \epsilon, A, X)$, where V and ϵ are the node and edge sets, X indicates the adjacency matrix and denotes the collection of nodes's features.

2.3. Feature extraction

The field of graph data representation has witnessed graph neural networks (GNNs) rising as a key area of intensive research (Zeng et al., 2021). Jiang et al. (2023) introduced the DropAGG mechanism for GNNs' layer-wise message passing. Expanding on the effectiveness of DropAGG, BADMDA enhances this aggregation approach by integrating

a bi-directional attention mechanism, enabling more effective feature extraction and improved adaptation to the heterogeneity of graph data.

Firstly, for any given node v with a feature vector x_v , To be specific, if $\delta_v = 1$, with a certain sampling ratio, BADMDA randomly samples nodes $\tilde{V}$ from the original node set V and conducts the aggregation operation only in the sampled node set $\tilde{V}$. Otherwise, it does not aggregate information from its neighbors. Typically, DropAGG executes the layer-based propagation process in the following steps:

$$\begin{aligned} \delta_v \in \{0, 1\} \text{ and } \delta_v \sim \text{Bernoulli}(1 - p), \\ \hat{X}_v^{(l+1)} \leftarrow \sigma(\alpha_{ij}^{(l)} \cdot \text{AGG}_{mean}\{A_{vu}, X_u^{(l)}, \forall u \in N(v) \cup v\}), \end{aligned} \quad (15)$$

where $p \in [0, 1]$ of the Bernoulli distribution in DropAGG serves to represent the drop rate. AGG_{mean} is defined as a mean aggregation function. $\sigma(\cdot)$ denotes activation function. $\alpha_{ij}^{(l)}$ symbolizes layer-wise weight parameters.

Subsequently, the attention coefficients between a node v and its adjacent node u using bi-directional attention are computed as follows:

$$\alpha_{ij}^{(l)} = \frac{\exp(\text{att}(x_v^{(l)}, x_u^{(l)}))}{\sum_{k \in N(v)} \exp(\text{att}(x_v^{(l)}, x_k^{(l)}))}, \quad (16)$$

$$\text{att}(x_v^{(l)}, x_u^{(l)}) = \text{LeakyReLU}(x_v^{(l)} W \cdot x_u^{(l)} + b),$$

where $\text{att}(x_v^{(l)}, x_u^{(l)})$ represents the attention score function between nodes v and u , W corresponds to the weights, and the term b corresponds to the bias. $N(v)$ encompasses all nodes adjacent to v . $\alpha_{ij}^{(l)}$ denotes normalized attention score (after **softmax**) indicating how much importance is given to node u when updating node v .

Finally, for networks that perform neighborhood aggregation, each layer broadens the scope of the influence dissemination by aggregating the neighborhoods of the previous layer. Implementing residual connection enables the aggregation of intermediate-layer representations to the ultimate layer, and this operation is described as:

$$\hat{X} = \sum_{i=1}^k w_i \hat{X}_v^{(i)}, \quad (17)$$

where w_i serves as the weight coefficient of layer i in feature aggregation step. It can adaptively adjust the influence of each nodes.

2.4. Prediction of potential associations

A decoder made up of a fully connected neural network is utilized to obtain the final prediction:

$$\hat{Z} = \text{sigmoid}(f(\hat{X}_m \oplus \hat{X}_d)), \quad (18)$$

To minimize the disparity between the true label and the predicted value, the cross-entropy loss function is applied in the subsequent step:

$$L_{ce} = - \sum_{(i,j) \in Z^+ \cup Z^-} [A_{ij} \log(\hat{Z}_{ij}) + (1 - A_{ij}) \log(1 - \hat{Z}_{ij})], \quad (19)$$

where A_{ij} stands for the true correlation between miRNA i and disease j , while $\hat{Z}_{ij}$ signifies the outcomes predicted by the BADMDA model. And Z^+ and Z^- denote the positive and negative samples, separately.

3. Experiments and results

3.1. Performance evaluation criteria

To comprehensively evaluate BADMDA's ability to forecast latent MDAs, benchmark datasets undergo 5-fold cross-validation (5-CV) experiments. The BADMDA's predictive performance is gauged by means of the area under the receiver operating characteristic curve (AUC) and the area under the Precision-Recall curve (AUPR). Additionally, to show BADMDA's merits from diverse aspects, metrics like accuracy (ACC),

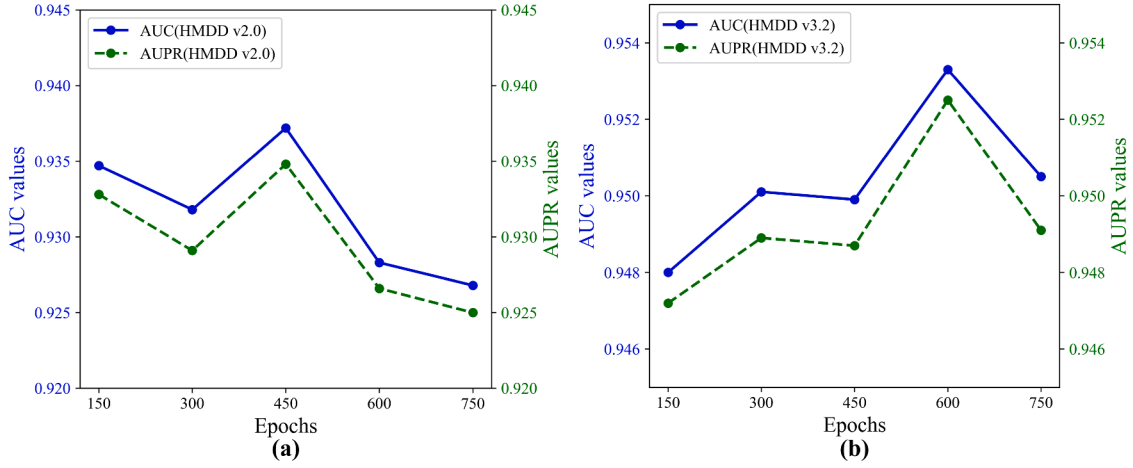


Fig. 2. Parameter analysis for epochs.

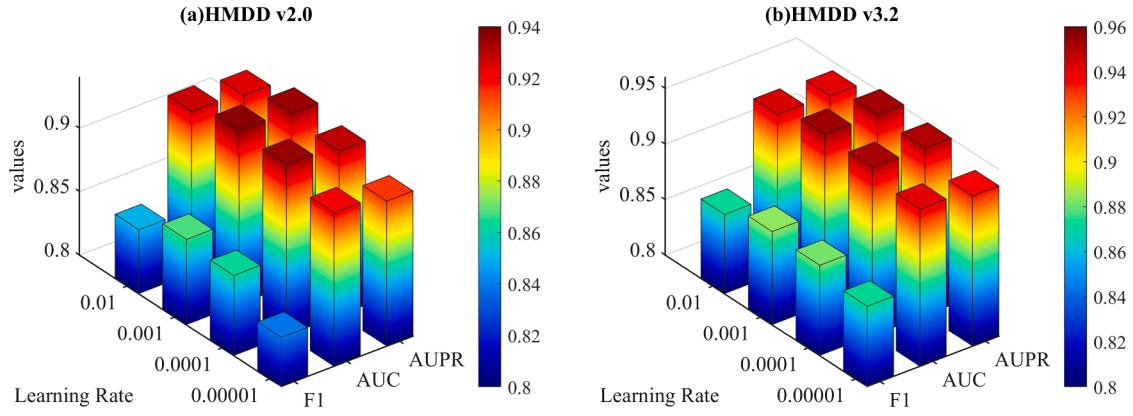


Fig. 3. Parameter analysis for learning rate.

F1-score (F1), precision (PRE), and recall are also considered.

$$\begin{aligned}
 FPR &= \frac{FP}{FP + TN}, \\
 TPR &= \frac{TP}{TP + FN}, \\
 ACC &= \frac{TP + TN}{TP + TN + FP + FN}, \\
 F1 &= 2 \times \frac{PRE \times Recall}{PRE + Recall}, \\
 PRE &= \frac{TP}{TP + FP}, \\
 Recall &= \frac{TP}{TP + FN},
 \end{aligned} \tag{20}$$

where TP is the number of positive samples that were correctly classified, and FP refers to positive samples that were classified incorrectly. Meanwhile, TN represents the number of negative samples identified correctly, with FN being the count of negative samples that were misclassified.

3.2. Parameter analysis

Table 1 presents the primary parameters and corresponding abbreviations employed in BADMDA. Before running the experiments, configuring parameters like training epochs n , learning rate lr , feature dimension d , and convolutional layers l serves as a key factor in optimizing BADMDA performance.

Table 2

Predictive performance of BADMDA and compared models on HMDD v2.0.

Models	AUC	AUPR	ACC	F1	PRE	Recall
MINIMDA	0.9178	0.9108	0.8435	0.8444	0.8404	0.8498
MDPBMP	0.9294	0.9281	0.8573	0.8590	0.8490	0.8692
AGAEMD	0.9164	0.9165	0.8386	0.8450	0.8130	0.8799
AMHMDA	0.9215	0.9186	0.8363	0.8402	0.8253	0.8601
DAEMDA	0.9318	0.9263	0.8550	0.8589	0.8373	0.8828
ESGC-MDA	0.9298	0.9242	0.8552	0.8565	0.8489	0.8645
PMDAGS	0.9222	0.9218	0.8464	0.8506	0.8283	0.8753
NIMCGCN	0.8062	0.7979	0.7368	0.7595	0.7003	0.8304
NIMGSA	0.8781	0.8786	0.7853	0.8014	0.7463	0.8659
VAEMDA	0.8969	0.9033	0.8238	0.8265	0.8145	0.8403
BADMDA	0.9372	0.9348	0.8665	0.8673	0.8619	0.8730

Note: Optimal results for each metric are bold.

The number of n significantly impacts BADMDA training performance. Fig. 2 indicates that the BADMDA's optimal value is achieved at 450 (HMDD v2.0) and 600 (HMDD v3.2).

The learning rate lr determines the magnitude of weight updates during network optimization. A too-high rate prevents convergence, while a too-low one slows it down. As Fig. 3 shows, setting lr to 0.001 maximizes the BADMDA's performance metrics.

The feature embedding size affects the dimension of its learnable parameter matrix, while the depth of feature spreading across nodes is affected by the number of GNN layers. More layers can aggregate higher-order neighbor features but may lose local structure. As Fig. 4 shows, BADMDA attains the highest AUC and AUPR scores at $d = 64$ and $l = 4$.

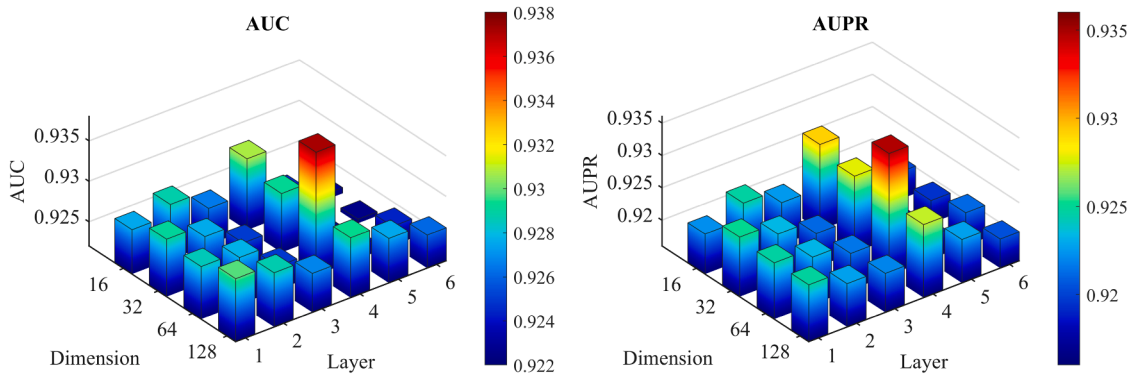


Fig. 4. AUC and AUPR at different dimension and number of layers.

Table 3

Predictive performance of BADMDA and compared models on HMDD v3.2.

Models	AUC	AUPR	ACC	F1	PRE	Recall
MINIMDA	0.9377	0.9345	0.8640	0.8671	0.8482	0.8869
MDPBMP	0.9378	0.9410	0.8705	0.8680	0.8856	0.8510
AGAEMD	0.9165	0.9273	0.8616	0.8596	0.8725	0.8471
AMHMDA	0.9398	0.9358	0.8575	0.8609	0.8529	0.8746
DAEMDA	0.9408	0.9395	0.8465	0.8393	0.8664	0.8331
ESGC-MDA	0.9488	0.9477	0.8788	0.8787	0.8797	0.8778
PMDAGS	0.9366	0.9359	0.8646	0.8678	0.8478	0.8888
NIMCGCN	0.7779	0.7683	0.6915	0.7287	0.6533	0.8278
NIMGSA	0.9301	0.9304	0.8533	0.8566	0.8390	0.8758
VAEMDA	0.9208	0.9234	0.8444	0.8471	0.8327	0.8623
BADMDA	0.9533	0.9525	0.8837	0.8840	0.8816	0.8863

Note: Optimal results for each metric are bold.

Table 4

The statistical test between BADMDA and the other ten models.

BADMDA vs	HMDD v2.0		HMDD v3.2	
	AUC p -values	AUPR p -values	AUC p -values	AUPR p -values
MINIMDA	1.5473×10^{-3}	1.6569×10^{-3}	9.7995×10^{-5}	8.4479×10^{-5}
MDPBMP	2.0064×10^{-2}	1.2304×10^{-2}	1.0050×10^{-4}	4.9040×10^{-4}
AGAEMD	3.1917×10^{-4}	1.0840×10^{-3}	9.7518×10^{-6}	1.1682×10^{-4}
AMHMDA	1.4107×10^{-3}	8.6520×10^{-3}	4.0133×10^{-4}	5.9686×10^{-4}
DAEMDA	3.8596×10^{-2}	3.7905×10^{-2}	3.8465×10^{-4}	8.1353×10^{-4}
ESGC-MDA	2.3750×10^{-2}	1.6661×10^{-2}	1.0421×10^{-2}	9.2155×10^{-3}
PMDAGS	1.9918×10^{-3}	1.6410×10^{-2}	7.4947×10^{-5}	1.1632×10^{-4}
NIMCGCN	3.9089×10^{-7}	1.9275×10^{-6}	6.3875×10^{-9}	7.8660×10^{-9}
NIMGSA	9.3596×10^{-6}	6.6675×10^{-5}	2.0433×10^{-5}	3.7461×10^{-5}
VAEMDA	4.2825×10^{-5}	6.4474×10^{-4}	5.3538×10^{-6}	1.2539×10^{-5}

3.3. Comparison with other models

BADMDA model's performance is measured by comparing it with 10 advanced models, which are tested on two benchmark datasets. For fairness considerations, all evaluated methods are tested on their respective benchmark datasets, strictly following the hyper-parameter configurations stated in their original publications. The comparison methods are as follows:

MINIMDA (Lou et al., 2022) predicts latent links by utilizing multi-modal networks and aggregating diverse neighborhood information.

MDPBMP (Yu et al., 2022) uses seven symmetrical meta-paths to take shape a heterogeneous miRNA-disease-gene network, updates node features, and combines the embeddings for measuring relevance scores.

AGAEMD (Zhang et al., 2022) employs a strategy that integrates node-level attention mechanisms into an AE framework for MDA prediction.

AMHMDA (Ning et al., 2023) fuses attention-aware multi-view similarity networks and hyper-graph learning, aiming to MDA identification.

DAEMDA (Dong et al., 2023) incorporates dual-channel attention encoding, leveraging both similarity networks and heterogeneous networks to predict MDAs.

ESGC-MDA (Bi et al., 2024) utilizes an enhanced simple graph convolution network to identify MDAs.

PMDAGS (Yan & Duan, 2024) forecasts MDAs by employing a graph non-linear diffusion convolution network.

NIMCGCN (Li et al., 2020) leverages neural inductive matrix completion in conjunction with GCN for inferring MDAs.

NIMGSA (Jin et al., 2022) exploits intricate relationships via the integration of graph AEs, self-attention, and neural inductive matrix completion.

VAEMDA (Zhang et al., 2019) introduces an unsupervised deep learning framework harnessing variational autoencoders (VAEs) to forecast candidate MDAs.

As listed in Table 2 on the HMDD v2.0 dataset, BADMDA achieved an AUC of 0.9372 and an AUPR of 0.9348. Compared with all baseline models, it consistently demonstrated improvements in both metrics: relative gains in AUC ranged from 0.58% over DAEMDA to 16.25% over NIMCGCN, while AUPR improvements ranged from 0.72% over MDPBMP to 17.16% over NIMCGCN. In Table 3 on the HMDD v3.2 dataset, BADMDA maintained its leading performance with an AUC of 0.9533 and an AUPR of 0.9525. Across all baselines, AUC improvements ranged from 0.47% over ESGC-MDA to 22.55% over NIMCGCN, and AUPR gains ranged from 0.51% over ESGC-MDA to 23.98% over NIMCGCN. These results quantitatively demonstrate that BADMDA consistently outperforms existing methods in terms of predictive accuracy and robustness across different datasets.

A paired t -test was conducted with a significance threshold set at $p' = 0.05$. If the calculated p -value exceeded this threshold, the null hypothesis was retained, indicating no significant difference between the two methods. As shown in Table 4, all calculated p -values were below the threshold, indicating that the differences are statistically significant. Specifically, BADMDA achieved higher AUC and AUPR values compared with all baseline methods on 5-CV, confirming its superior predictive performance in MDA prediction as supported by both the statistical test and the quantitative evaluation metrics.

3.4. Ablation experiments

To assess each module's contribution to BADMDA's performance, several model variants were created and evaluated via 5-CV. Their configurations are as follows:

(1) w/o A: Represents CKA-MKL without adaptive weights, directly integrating multi-source features via average fusion.

(2) w/o B: Retains the core strategy of DropAGG strategy, but removes bi-directional attention.

(3) w/o C: Completely removes the DropAGG strategy from the message passing process in the bi-directional attention network.

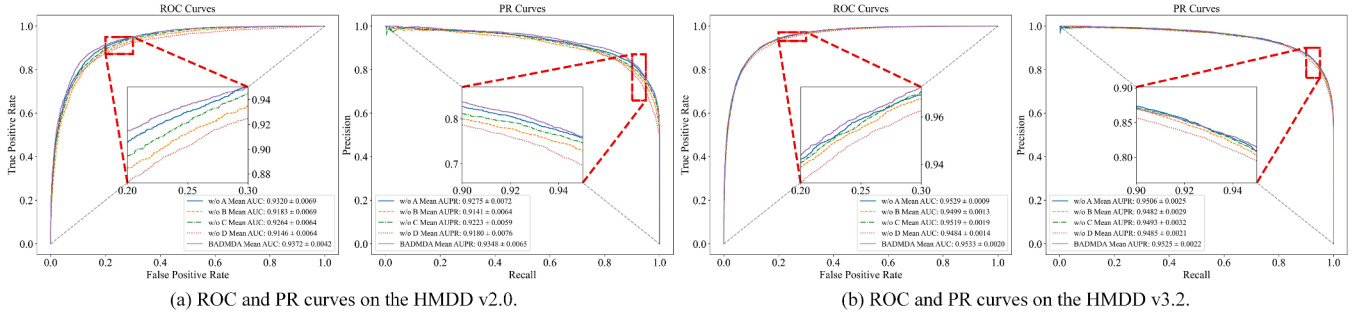


Fig. 5. ROC and PR curves for different modules on benchmark datasets.

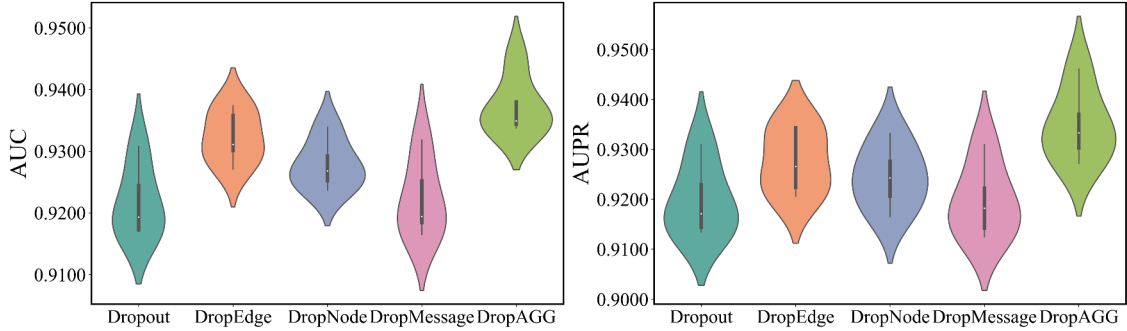


Fig. 6. Comparison of dropping strategies on AUC and AUPR.

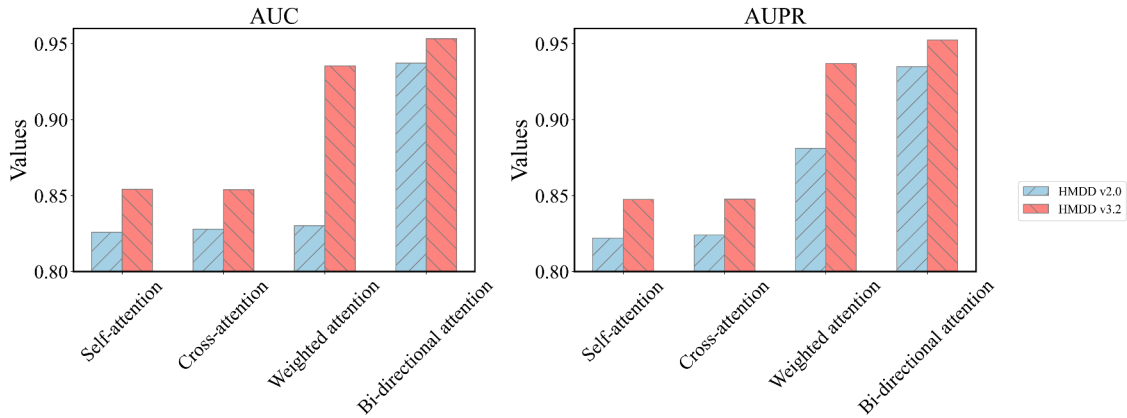


Fig. 7. Comparison of various attention mechanisms on AUC and AUPR.

(4) w/o D: Simplifies BADMDA by removing modules A, B, and C simultaneously.

The variants' ROC and PR curves on two benchmark datasets displayed in Fig. 5 (a) and (b). Removing any single module (A, B, or C) reduces BADMDA's performance, while eliminating all three causes the steepest decline, highlighting their essential and complementary functions for the BADMDA's effectiveness.

The impact of various random dropping strategies, including Dropout, DropEdge, DropNode, DropMessage, and DropAGG, on the message propagation process was assessed. From the depiction in Fig. 6, it can be seen that DropAGG performs most favorably in relation to AUC and AUPR.

To further explore BADMDA's capabilities in the bi-directional attention framework, experimental analyses with self-attention, cross-attention, weighted attention, and bi-directional attention mechanisms were conducted. As shown in Fig. 7, the bi-directional attention mechanism achieved the best results.

3.5. Case studies

To showcase the predictive power of identifying disease-associated miRNAs, comprehensive case studies focusing on lung cancer and hepatocellular carcinoma (HCC) were conducted. On a global scale, lung cancer ranks as the "top killer" in the oncology field, with its morbidity and mortality long remaining the highest among all cancers. Over ten consecutive years, it has continued to be the primary driver of cancer deaths worldwide (Luo et al., 2025). This grim epidemiological finding not only underscores the urgent demand for advanced methods to address lung cancer but also elevates its status as a key case study in predicting disease-associated miRNAs. As the most prevalent type of liver cancer, HCC accounts for approximately 75–90% of liver cancer cases (Younossi & Henry, 2021). Due to its lack of early symptoms, HCC often leads to late diagnosis and poor prognosis; thus, leveraging MDA predictions for early detection can facilitate the timely prevention and treatment of this malignancy.

Table 5

Top 15 miRNAs association with lung neoplasms on HMDD v3.2.

Rank	MiRNA	Evidence
1	hsa-mir-34b	D, H, 23,314,612
2	hsa-mir-25	D, H, 31208279, 30,919,763
3	hsa-mir-29c	D, H, 35,585,716
4	hsa-mir-100	D, H, 35307327, 32,364,673
5	hsa-mir-18a	D, H, 20,508,945
6	hsa-mir-140	D, H, 39,348,965
7	hsa-mir-200c	D, H, 36,587,786
8	hsa-mir-10a	D, H, 28,781,815
9	hsa-mir-1471	Unconfirmed
10	hsa-mir-494	D, H, 39016137, 37,480,691
11	hsa-mir-101-2	H, 21,849,855
12	hsa-mir-181a-2	D, H, 20,363,096
13	hsa-mir-199b	D, H, 21,563,230
14	hsa-mir-32	D, H, 22349819, 17,028,596
15	hsa-mir-101-1	H, 21849855, 16,530,703

Table 6

Top 15 miRNAs association with Hepatocellular carcinoma on HMDD v3.2.

Rank	MiRNA	Evidence
1	hsa-mir-148b	D, H, 36,096,296
2	hsa-mir-151a	D, H, 38102691, 30,781,550
3	hsa-mir-503	D, H, 33,398,379
4	hsa-mir-574	D, H, 29,065,698
5	hsa-mir-381	D, H, 26,677,080
6	hsa-mir-26a-2	D, H, 32585630, 36,381,584
7	hsa-mir-449a	D, H, 22,641,068
8	hsa-mir-638	D, H, 36424264, 33,426,736
9	hsa-mir-18b	D, H, 30,272,372
10	hsa-mir-206	D, H, 34,706,869
11	hsa-mir-185	D, H, 31658041, 28,240,051
12	hsa-mir-373	D, H, 36708579, 33,361,718
13	hsa-mir-338	D, H, 33416148, 33,028,110
14	hsa-mir-451a	D, H, 33658044, 33,206,588
15	hsa-mir-134	D, H, 35116608, 32,711,110

In the case studies, the core goal was to identify potential correlation with designated disease. Accordingly, the full collection of documented relationships within HMDD V3.2 was initially adopted as the training dataset for predictive tasks. Next, known disease-related miRNAs linked to lung cancer and HCC were excluded from this training dataset; subsequently, features of lung cancer and HCC were fed into the BADMDA model to calculate association scores between these two diseases and 788 miRNAs. Thereafter, these miRNAs were sorted in descending order according to the association scores, yielding the top 15 miRNAs associated with lung cancer and HCC. Finally, the ranking outcomes of these miRNAs were verified via relevant databases and academic literature. Validation uses the authoritative dbDEMC (Xu et al., 2022) (abbreviated "D"), HMDD v4.0 (Cui et al., 2024) (abbreviated "H") and relevant scientific literature. The evidence numbers are PubMed Unique Identifier (PMID).

As tabulated in Table 5, 14 out of the 15 top-ranked predicted miRNAs have been verified by at least one of the following: the dbDEMC database, the HMDD v4.0 repository, or existing scholarly literature. An even more compelling validation emerges from Table 6, where all these miRNAs have been independently corroborated by dbDEMC, HMDD v4.0, and relevant research publications. This triple-layer confirmation not only attests to the reliability of the prediction model but also underscores the consistency of the findings across multiple authoritative sources. The results of these case studies can provide biologists with potential disease-related candidate miRNAs, which not only offers new targets for exploring the molecular mechanisms of lung cancer and HCC but also further demonstrates the practical value of the BADMDA model in mining miRNAs associated with specific diseases, thus facilitating wet experiments to a certain extent.

4. Conclusion

The BADMDA model proposed in this study innovatively integrates a bi-directional attention network with the DropAGG strategy. This new framework overcomes the limitations of existing MDA prediction methods. Based on comprehensive and stringent assessments, it has been demonstrated that the BADMDA model surpasses ten advanced models across a wide array of evaluation metrics. Case analyses on lung cancer and hepatocellular carcinoma further confirm its robustness and superiority in MDA prediction, providing insights for disease diagnosis and treatment.

Nonetheless, there remain some deficiencies in BADMDA. Integrating prior information could offer more comprehensive features, which is a key area for improvement. In future research, the integration of diverse biological information sources, including drugs, lncRNAs, circRNAs, and genes related to miRNA/disease, will be contemplated to further enhance the prediction performance.

CRedit authorship contribution statement

Yan-Fang Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization; **Yue Gao:** Validation, Formal analysis, Data curation; **Ming-Li Cui:** Validation, Data curation; **Ying-Lian Gao:** Writing – review & editing, Project administration, Methodology; **Yan-Li Wang:** Writing – review & editing, Validation, Conceptualization; **Jin-Xing Liu:** Writing – review & editing, Methodology, Conceptualization.

Data availability

The source code and data are available at <https://github.com/011211/BADMDA>. The experiments for HGATMM were conducted in the following computing and hardware environment: the software ran on a Windows 11 operating system, with Python 3.7.5 as the programming language and PyTorch 1.12.1 as the deep learning framework; the hardware configuration included an Intel Core i5-13500H processor, 16 GB of RAM, and an NVIDIA RTX 4050 graphics card.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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